

QRNG comparison report

stagea_ibm_marrakesh_20260714-210803_processed.txt vs optimal_harvested_ibm_marrakesh_20260714-214816_
Generated 2026-07-15 09:15

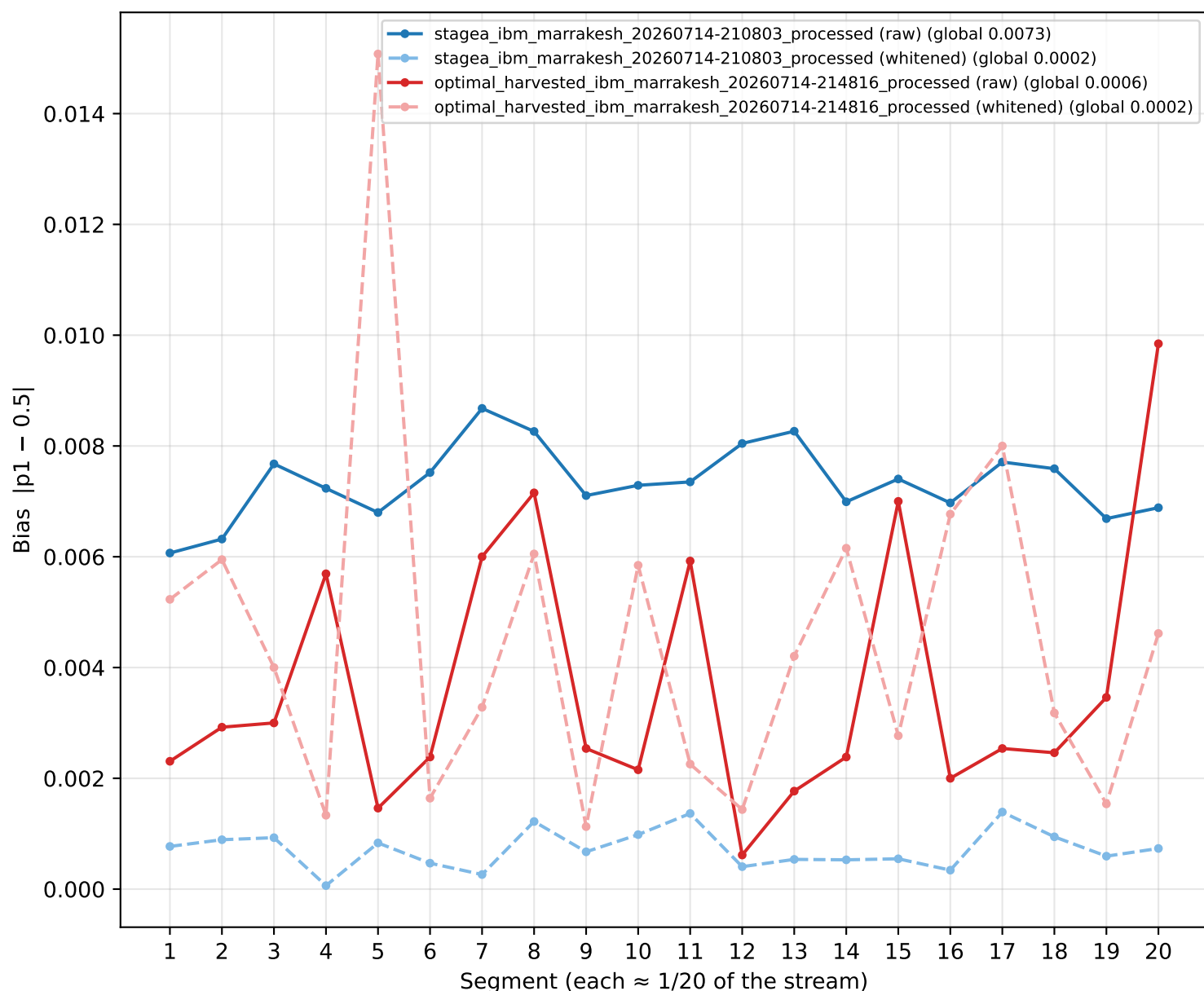
Bits used per file: ALL | NIST bits/partition: 1,000,000 | ML windows [8, 16, 24], 8 folds | Markov orders [1, 2, 4]

stagea_ibm_marrakesh_20260714-210803_processed.txt	stagea_ibm_marrakesh_20260714-210803_processed.txt	optimal_harvested_ibm_marrakesh_20260714-214816_	optimal_harvested_ibm_marrakesh_20260714-214816_
Total bits	10,140,000	7,605,000	260,000
Global bias p1-0.5	0.0073	0.0002	0.0006
Min-entropy / bit	0.9790	0.9994	0.9983
NIST pass rate	0.48	1.00	1.00
NIST passed / run	29/60	66/66	57/57
Next-bit verdict	PASS	PASS	PASS
Next-bit worst AUC	0.5000	0.5000	0.5000
Markov verdict	PASS	PASS	PASS
Markov max excess	0.0007	0.0000	0.0000

All four streams side by side: each input file contributes its RAW stream and its WHITENED (SHA-512, 75% kept) stream. Lower bias is better; min-entropy near 1.0 is better; NIST pass rate near 1.0 is better; next-bit / Markov verdict PASS means no detectable structure (accuracy ~0.5 with p >= 0.01 and a 95% CI that includes 0.5).

Bias per segment (20 splits)

Local balance along each stream — all four streams overlaid



Each stream is cut into 20 equal segments; the plotted value is how far that segment's fraction of ones sits from 0.5. Solid lines = raw, dashed = whitened. A good source is LOW and FLAT: no segment drifts and there is no rising/falling trend. Whitened lines should sit near the bottom regardless of how their raw counterpart behaves — that is the extractor removing residual imbalance. Reading tip: compare the two SOLID (raw) lines to judge the hardware; the dashed lines show what post-processing recovers.

Bias per segment — data ($|p1 - 0.5|$ per split)

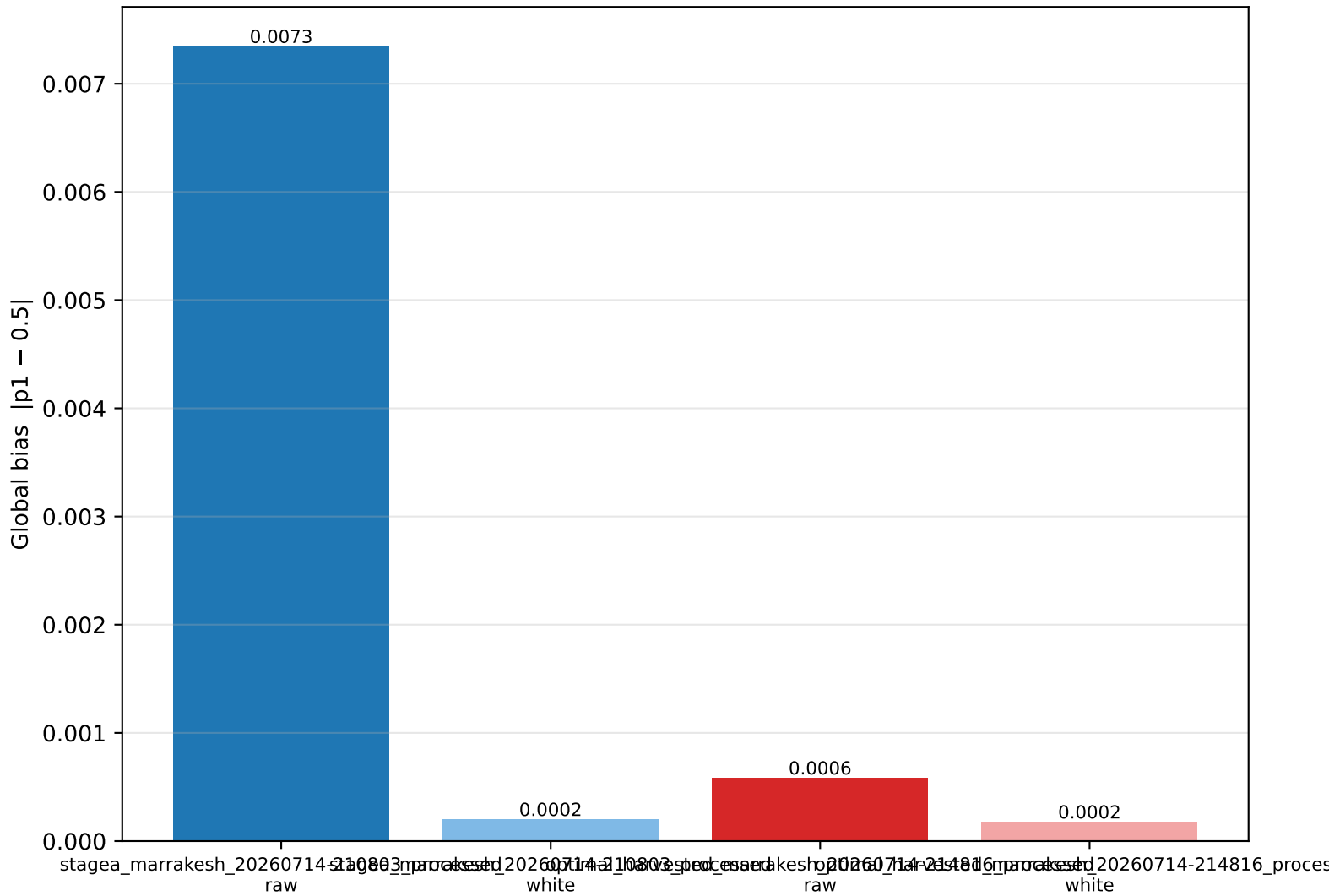
Data table (same numbers as the chart)

Split	sea_marrakesh_20260714-214	sea_marrakesh_20260714-214	sea_marrakesh_20260714-214	sea_marrakesh_20260714-214
1	0.0061	0.0008	0.0023	0.0052
2	0.0063	0.0009	0.0029	0.0059
3	0.0077	0.0009	0.0030	0.0040
4	0.0072	0.0001	0.0057	0.0013
5	0.0068	0.0008	0.0015	0.0151
6	0.0075	0.0005	0.0024	0.0016
7	0.0087	0.0003	0.0060	0.0033
8	0.0083	0.0012	0.0072	0.0061
9	0.0071	0.0007	0.0025	0.0011
10	0.0073	0.0010	0.0022	0.0058
11	0.0074	0.0014	0.0059	0.0023
12	0.0080	0.0004	0.0006	0.0014
13	0.0083	0.0005	0.0018	0.0042
14	0.0070	0.0005	0.0024	0.0062
15	0.0074	0.0005	0.0070	0.0028
16	0.0070	0.0003	0.0020	0.0068
17	0.0077	0.0014	0.0025	0.0080
18	0.0076	0.0009	0.0025	0.0032
19	0.0067	0.0006	0.0035	0.0015
20	0.0069	0.0007	0.0098	0.0046

Same values as the chart above. Lower is better; look for any column that is systematically larger (a more biased stream) or any single spiking row (localised imbalance).

Global bias & min-entropy — raw vs whitened

Whole-stream imbalance for all four streams



Whole-stream bias (one number per stream). The whitened bars should be far lower than their raw counterparts — direct evidence the SHA-512 extractor removes imbalance while keeping 75% of the data. Min-entropy per bit (table) converts bias to 'how many bits of true randomness per output bit'; 1.0 is ideal.

Global bias & min-entropy — data

Data table (same numbers as the chart)

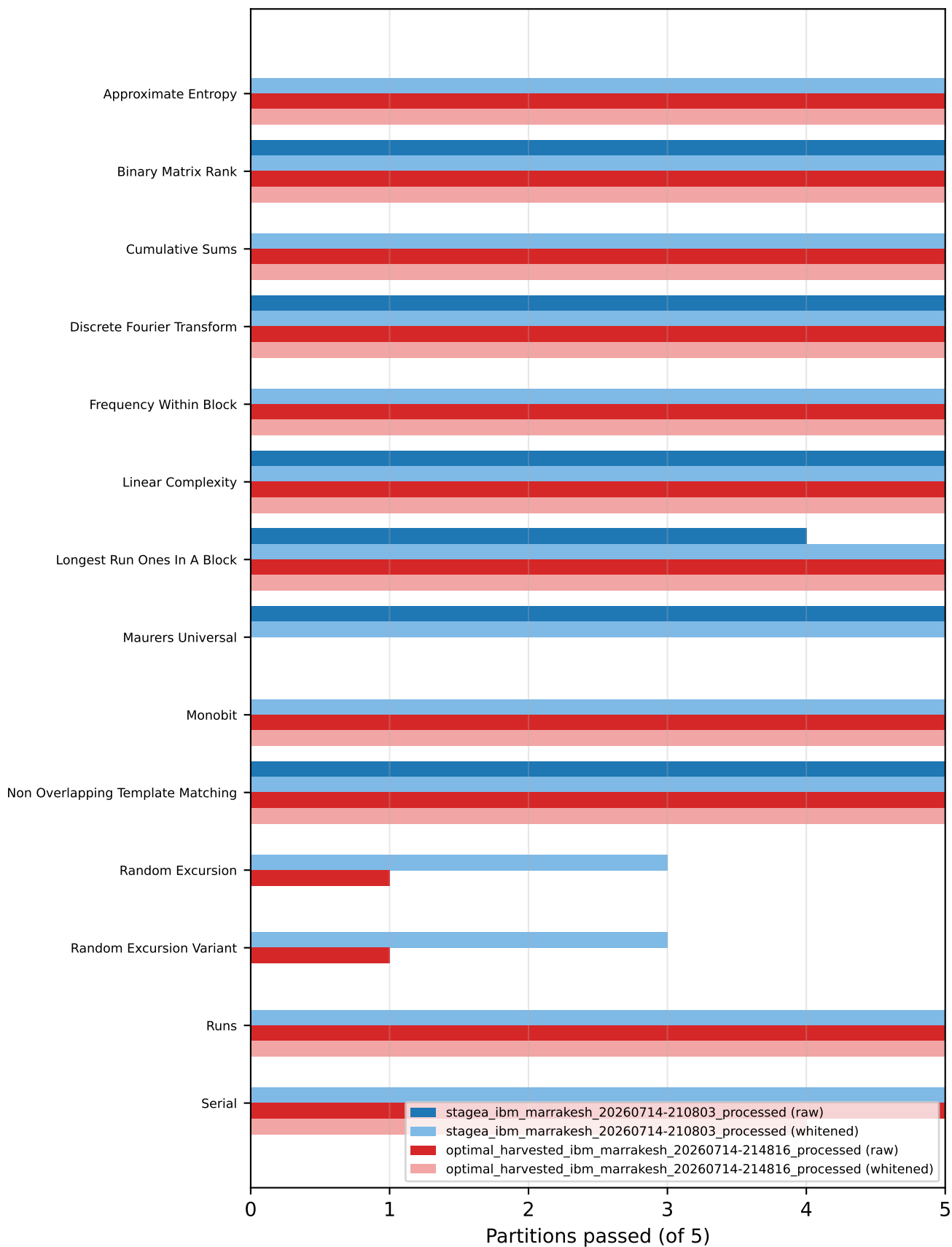
Marrakesh_20260714-214	Marrakesh_20260714-214	Marrakesh_20260714-214	Marrakesh_20260714-214	Marrakesh_20260714-214
Global bias	0.0073	0.0002	0.0006	0.0002
Min-entropy/bit	0.9790	0.9994	0.9983	0.9995
Total bits	10,140,000	7,605,000	260,000	195,000

Same numbers as the chart. Compare each raw stream with its whitened version to quantify how much structure the extractor removed.

NIST SP 800-22 — partitions passed per test

Out of 5 partitions; all four streams grouped per test

Overall pass rate — stagea_marrakesh_20260714-210803_processed raw: 0.48 stagea_marrakesh_20260714-210803_processed white: 1.00 optim



The full NIST SP 800-22 battery runs on 5 partitions of each stream; a bar of 5 means the test passed in every partition ($p > 0.01$ each). Longer is better. Tests that need more bits than a partition provides are skipped by the library (so the test list can differ between streams if bit counts differ). Whitened streams should reach 5 across the board.

NIST SP 800-22 — data (partitions passed / run)

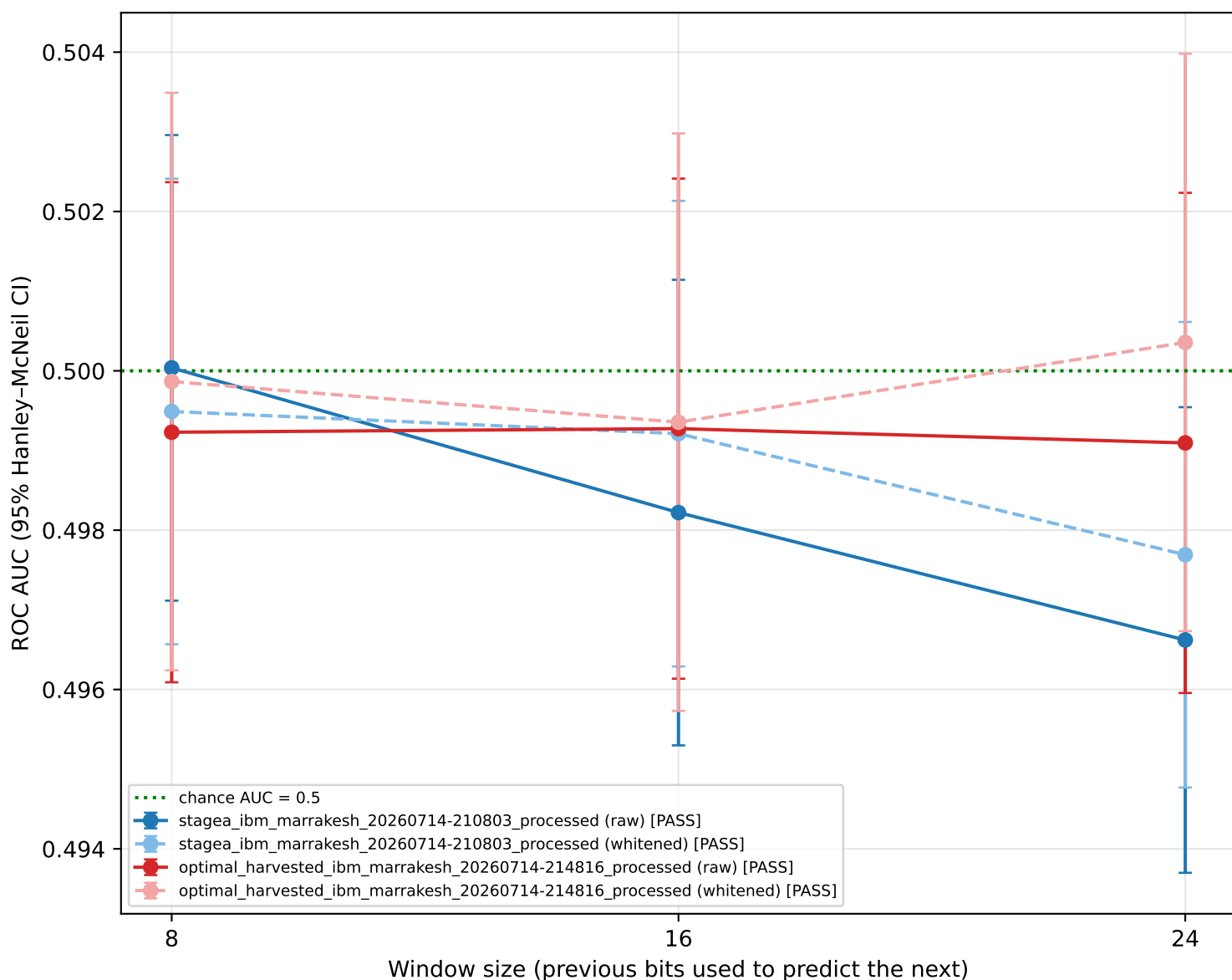
Data table (same numbers as the chart)

	NIST test	tegea_marrakesh_20260714-214803	tegea_marrakesh_20260714-214803	tegea_marrakesh_20260714-214803	tegea_marrakesh_20260714-214803
	Approximate Entropy	0/5	5/5	5/5	5/5
	Binary Matrix Rank	5/5	5/5	5/5	5/5
	Cumulative Sums	0/5	5/5	5/5	5/5
	Discrete Fourier Transform	5/5	5/5	5/5	5/5
	Frequency Within Block	0/5	5/5	5/5	5/5
	Linear Complexity	5/5	5/5	5/5	5/5
	Longest Run Ones In A Block	4/5	5/5	5/5	5/5
	Maurers Universal	5/5	5/5	—	—
	Monobit	0/5	5/5	5/5	5/5
Non	Overlapping Template Match	5/5	5/5	5/5	5/5
	Random Excursion	—	3/3	1/1	—
	Random Excursion Variant	—	3/3	1/1	—
	Runs	0/5	5/5	5/5	5/5
	Serial	0/5	5/5	5/5	4/5
	OVERALL pass rate	0.48	1.00	1.00	0.98

Same values as the chart. '4/5' means the test passed in 4 of the 5 partitions. '—' means the test was not eligible for that stream's length.

Next-bit predictability — strict, bias-robust (AUC)

AUC vs window size; 0.5 = no learnable structure



A logistic-regression model predicts each bit from the previous W bits, 8-fold time-ordered CV, for W in $[8, 16, 24]$. We use ROC AUC as the primary signal because it is BIAS-ROBUST: a biased-but-independent stream still gives $AUC \sim 0.5$, whereas plain accuracy would rise just from guessing the majority bit. Structure is flagged only if a window's AUC 95% CI lies ABOVE 0.5 (or accuracy beats the majority baseline). Verdict in the legend is the strict AND over all windows. (Raw bias itself is reported separately in the bias analysis.)

Next-bit predictability — data (per stream × window)

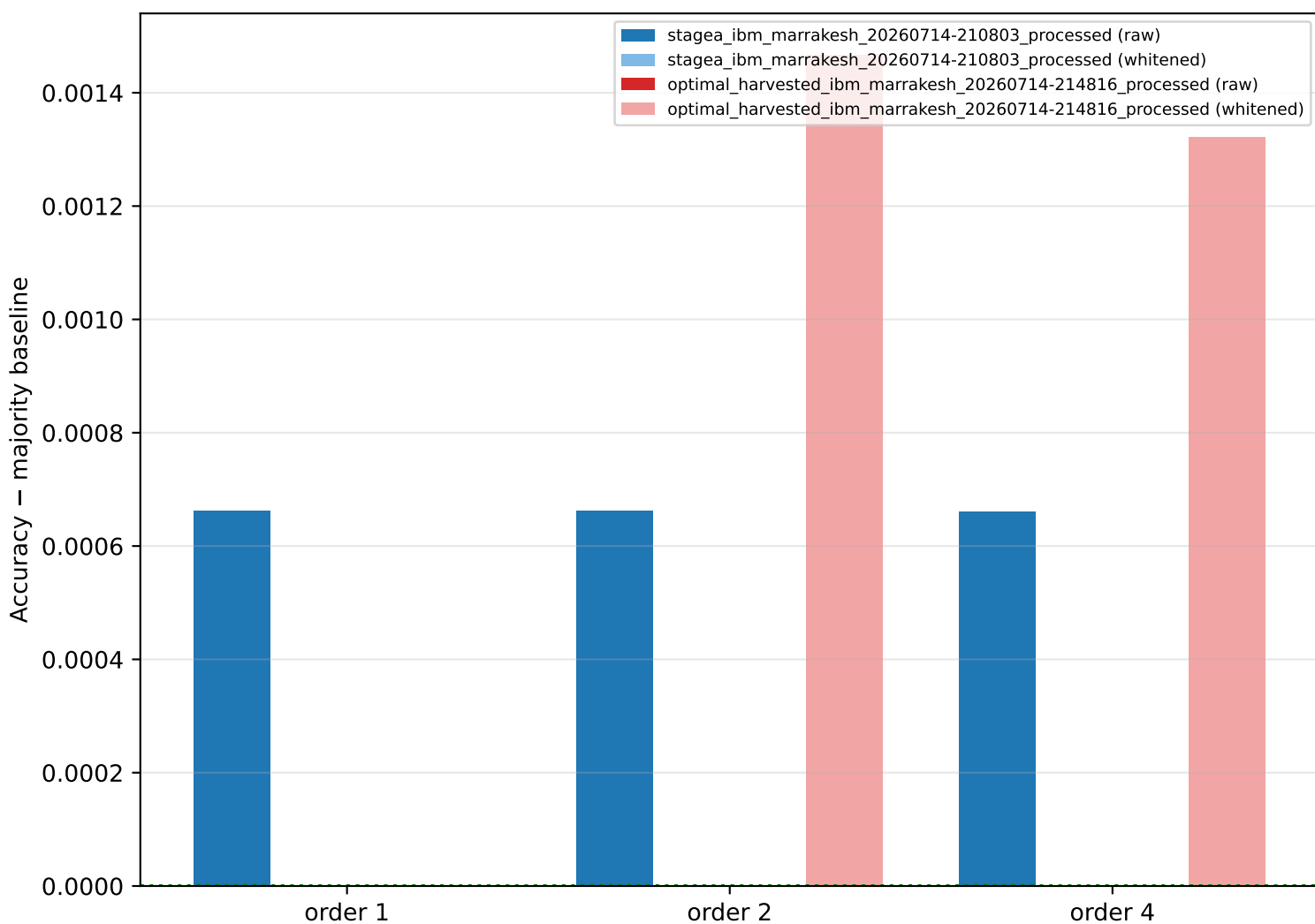
Data table (same numbers as the chart)

	Stream	Window	Accuracy	Majority base	AUC	AUC 95% CI	p	pass
rrakesh	_20260714-2108	8	0.5052	0.5054	0.5000	[0.497,0.503]	0.000	PASS
rrakesh	_20260714-2108	16	0.5052	0.5054	0.4982	[0.495,0.501]	0.000	PASS
rrakesh	_20260714-2108	24	0.5016	0.5054	0.4966	[0.494,0.500]	0.113	PASS
rrakesh	_20260714-21080	8	0.5004	0.5009	0.4995	[0.497,0.502]	0.391	PASS
rrakesh	_20260714-21080	16	0.5003	0.5009	0.4992	[0.496,0.502]	0.421	PASS
rrakesh	_20260714-21080	24	0.4985	0.5009	0.4977	[0.495,0.501]	0.870	PASS
ed_marrakesh	_20260714	8	0.4999	0.5006	0.4992	[0.496,0.502]	0.532	PASS
ed_marrakesh	_20260714	16	0.4999	0.5006	0.4993	[0.496,0.502]	0.536	PASS
ed_marrakesh	_20260714	24	0.5000	0.5006	0.4991	[0.496,0.502]	0.506	PASS
ed_marrakesh	_20260714-	8	0.4978	0.5002	0.4999	[0.496,0.503]	0.916	PASS
ed_marrakesh	_20260714-	16	0.4982	0.5002	0.4994	[0.496,0.503]	0.864	PASS
ed_marrakesh	_20260714-	24	0.4991	0.5002	0.5004	[0.497,0.504]	0.715	PASS

'Majority base' = accuracy obtainable by always guessing the more common bit (pure bias). A window PASSES when AUC's CI includes/sits at 0.5 AND accuracy does not significantly exceed the majority baseline.

Markov dependency — multi-order (bias-robust)

Accuracy ABOVE the majority baseline; 0 = no dependency



A k-th order Markov predictor guesses the next bit from the previous k bits. We plot how far its accuracy exceeds the MAJORITY (bias-only) baseline, so pure bias reads as ~0 — only genuine short-range DEPENDENCY lifts a bar above the green line. Whitened streams should sit at 0 at every order. (Overall bias is reported separately.)

Markov dependency — data (accuracy, baseline, excess per order)

Data table (same numbers as the chart)

	Stream	o1 acc	o2 acc	o4 acc	base	o1 excess	o2 excess	o4 excess	verdict
rrakesh_	20260714-2108	0.5069	0.5069	0.5069	0.5063	0.0007	0.0007	0.0007	PASS
rrakesh_	20260714-2108	0.4993	0.4998	0.5000	0.5004	-0.0011	-0.0006	-0.0004	PASS
ed_marra	kesh_20260714	0.4996	0.4996	0.5003	0.5006	-0.0010	-0.0010	-0.0003	PASS
d_marra	kesh_20260714	0.5000	0.5016	0.5015	0.5002	-0.0002	0.0015	0.0013	PASS

'base' = majority baseline (pure bias). 'excess' = accuracy – baseline; a positive, statistically significant excess at any order flags dependency and fails the Markov verdict.

Conclusions

Strict per-stream verdict and raw-hardware comparison

Stream	Bias	NIST	Next-bit	Markov	OVERALL
marrakesh_20260714-21080	0.0073	low	PASS	PASS	FAIL
marrakesh_20260714-210803	0.0002	OK	PASS	PASS	PASS
ed_marrakesh_20260714-	0.0006	OK	PASS	PASS	PASS
ed_marrakesh_20260714-2	0.0002	OK	PASS	PASS	PASS

Hardware (raw) comparison: lower global bias = optimal_harvested_ibm_marrakesh_20260714-214816_processed (raw) (0.0006 vs 0.0073).

Strict next-bit = multi-window logistic-regression CV with Wilson CIs; a single predictable window fails the stream. Markov spans orders [1, 2, 4].

Note: whitened streams typically PASS everything — a good extractor whitens any high-entropy source — so judge the HARDWARE on the RAW rows, and read the whitened rows as 'what post-processing delivers'.

OVERALL PASS requires: NIST pass rate ≥ 0.90 , next-bit verdict PASS, and Markov verdict PASS. This is a guide, not a formal certification; sample size and calibration still matter.